

Student Grade Tracker

AI Pair-Programmed REST API Project

Name: Sai Ankith Reddy Kancherla **Course:** SDE AI — IIT Mandi (Himshikhar Program)
Date: June 15, 2026

Problem Statement

Managing student grades manually is inefficient and error-prone. There is no easy way to:

- Add and store student grade records
- Automatically calculate GPA
- Search or delete student records quickly

A simple, accessible web-based solution is needed.

Approach / Methodology

1. Designed a REST API with 4 endpoints using FastAPI (Python)
2. Built a clean frontend website in HTML/CSS/JS to interact with the API
3. Used AI pair programming (Claude) to accelerate development
4. Wrote automated tests using Pytest to verify correctness
5. Documented the entire AI collaboration process

Technologies Used

Technology	Purpose
Python 3.14	Backend language
FastAPI	REST API framework
Uvicorn	Server to run the API
HTML/CSS/JS	Frontend website
Pytest	Automated testing

Technology	Purpose
Claude AI	AI pair programming assistant

Key Features

- Add students with name, subject and grades
 - Automatic GPA calculation
 - Search student by ID
 - Delete student records
 - Color coded GPA badges (green/yellow/red)
 - 5 automated tests — all passing
 - Interactive Swagger UI documentation at /docs
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Results / Output

- Fully functional REST API with 4 endpoints
- Clean dark themed frontend website
- 5/5 tests passing
- AI documentation showing collaboration process
- Live demo included